Association Between County-Level Socioeconomic Factors and Late-Stage Breast Cancer Among Women Under 50 in California

Abstract

Breast cancer remains a major public health concern, and early detection is critical for improving survival outcomes. However, socioeconomic conditions may influence access to screening and healthcare services, potentially affecting the stage at which breast cancer is diagnosed. This study investigates how county-level socioeconomic factors are associated with late-stage breast cancer rates among women under 50 in California.

Using county-level health and socioeconomic data, exploratory data analysis (EDA) was conducted, including correlation analysis and visualizations such as heatmaps and scatter plots. The analysis examines relationships between variables such as poverty rate, median income, and insurance coverage and the proportion of late-stage breast cancer diagnoses. The findings suggest that counties with higher poverty and uninsured rates tend to show higher percentages of late-stage diagnoses, highlighting the potential role of socioeconomic disparities in breast cancer outcomes.

Introduction

Background

Social determinants of health are the economic and social conditions that influence health outcomes. Factors such as income, poverty, and access to health insurance can affect an individual's ability to receive medical care, preventive screenings, and timely treatment.

For breast cancer, early detection through screening greatly improves treatment outcomes. However, people living in areas with higher poverty or limited healthcare access may experience delays in diagnosis, leading to higher rates of late-stage detection.

This study focuses on women under 50 because breast cancer in younger women is often diagnosed at a later stage and can be more aggressive. Examining the relationship between socioeconomic factors and breast cancer outcomes can help identify potential health disparities and inform efforts to improve early detection and access to care.

Research Question

This study aims to answer the following research question:

How are county-level socioeconomic factors associated with late-stage breast cancer rates among women under 50 in California?

Specifically, the analysis explores whether factors such as poverty rate, median income, and insurance coverage show patterns or associations with the proportion of breast cancer cases diagnosed at a late stage.

Dataset Overview

This analysis uses four datasets containing county-level information on breast cancer outcomes and socioeconomic conditions in California. The breast cancer incidence and late-stage diagnosis rates were obtained from **State Cancer Profiles (statecancerprofiles.cancer.gov)**, which compiles data from **State Cancer Registries**. The socioeconomic variables, including median household income, poverty rate, and health insurance coverage, were obtained from the **U.S. Census Bureau's American Community Survey (ACS)**.

Because these variables were stored in separate datasets, the relevant variables were first selected and then merged into one dataset using the county **FIPS code** as the common identifier. The final merged dataset includes variables such as late-stage breast cancer rate, median household income, poverty rate, and percent uninsured for counties in California.

Data Cleaning Process

Before conducting the analysis, the datasets were cleaned and prepared to ensure consistency and accuracy. During the cleaning process, variable formats were checked and standardized, and missing or inconsistent values were reviewed to ensure the dataset was suitable for analysis.

In addition, counties were categorized as rural or urban to explore potential differences in cancer outcomes by geographic area. Several rural counties had suppressed cancer counts due to small populations, resulting in missing values for some variables. These counties were retained in the dataset whenever possible and were only excluded when necessary for specific statistical tests that required complete observations. This limitation should be considered when interpreting comparisons between rural and urban counties.

Statistical Methods Used

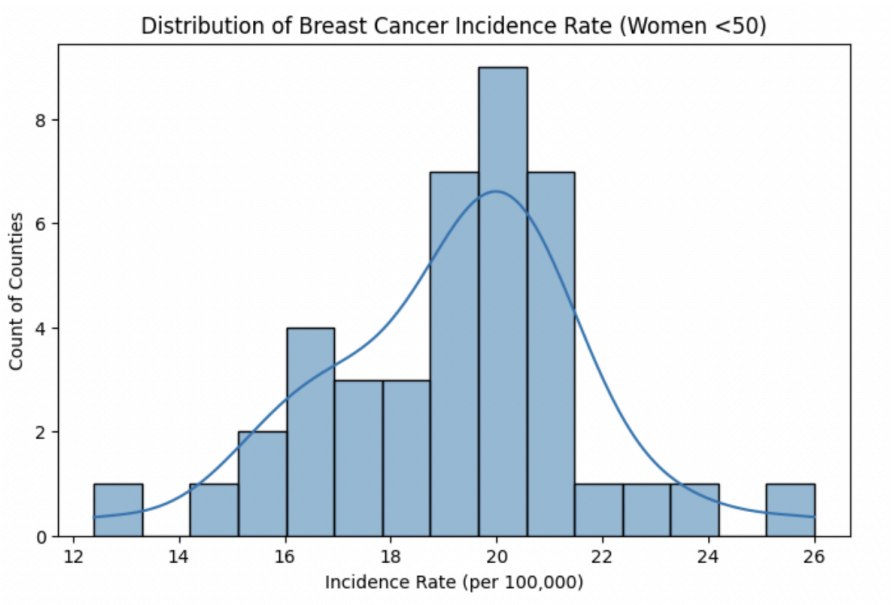
Exploratory data analysis (EDA) was conducted to examine the data and identify patterns between variables. Histograms were used to visualize the distribution of individual variables for univariate analysis, helping to better understand the data and identify potential outliers.

Scatter plots were then used to explore relationships between socioeconomic factors and late-stage breast cancer rates. A correlation matrix and heatmap were also created to examine the strength and direction of relationships among the variables.

In addition, linear regression analysis was conducted to evaluate whether poverty rate, median income, and uninsured rate could help explain differences in late-stage breast cancer rates across counties.

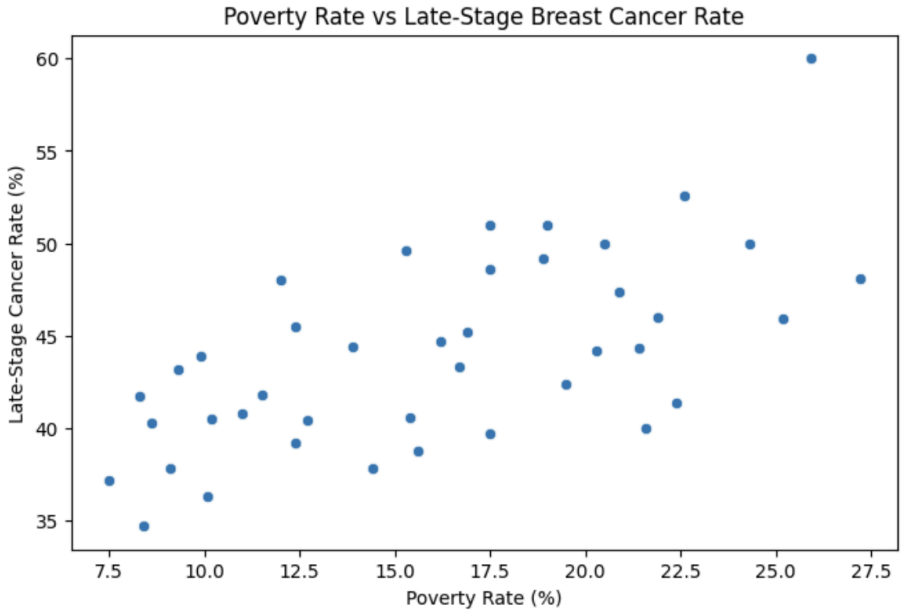
Key Findings & Visualizations

To understand the overall distribution of breast cancer incidence rates among women under 50, a histogram was created. The distribution shows that most counties have similar breast cancer incidence rates among women under 50, generally around 18–21 cases per 100,000.

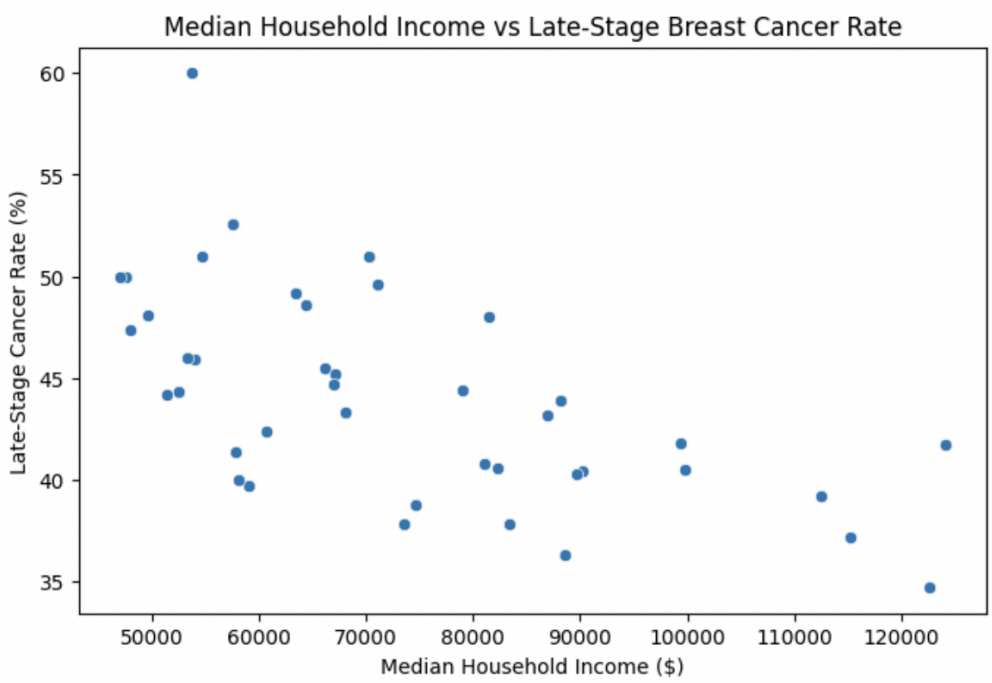


The analysis below shows several relationships between socioeconomic factors and late-stage breast cancer rates.

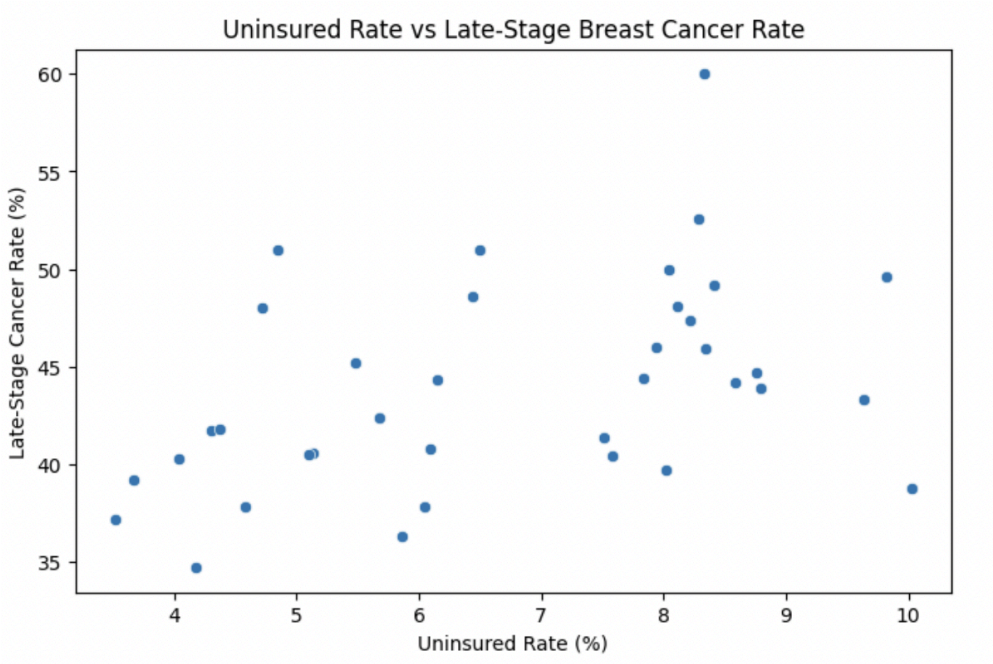
The following scatter plot indicates that counties with higher poverty rates tend to have higher late-stage breast cancer rates, suggesting that economic disadvantage may be linked to delayed diagnosis.



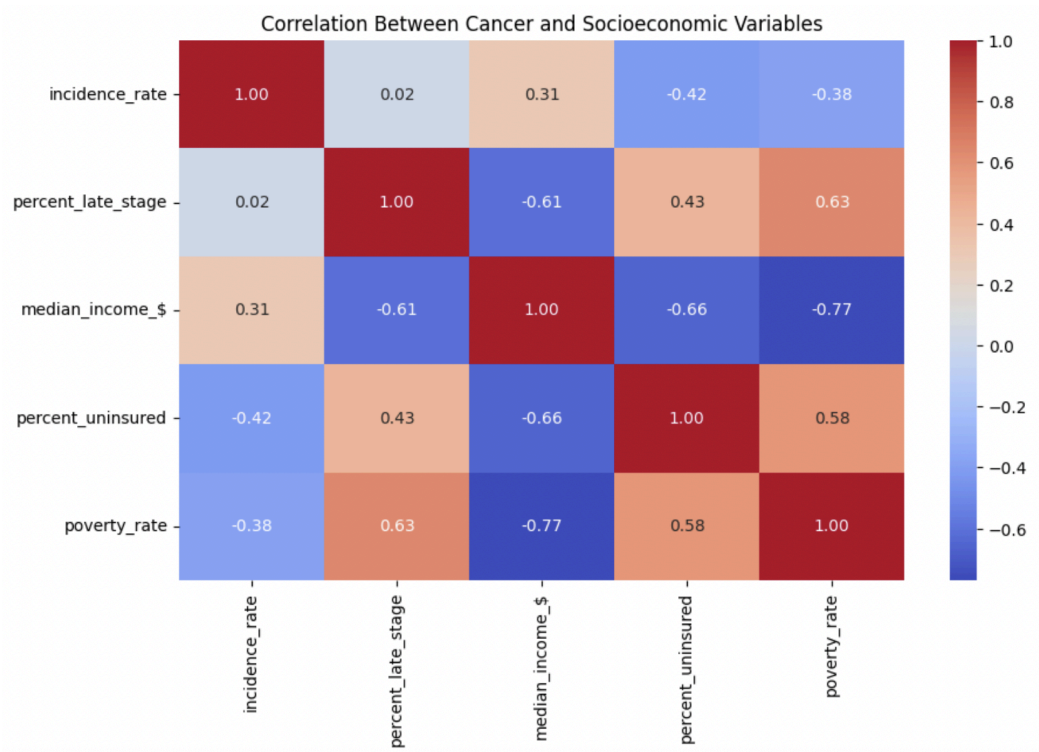
Median household income shows the opposite pattern. Counties with higher income levels generally have lower percentages of late-stage cancer diagnoses, which may reflect differences in healthcare access and preventive screening.



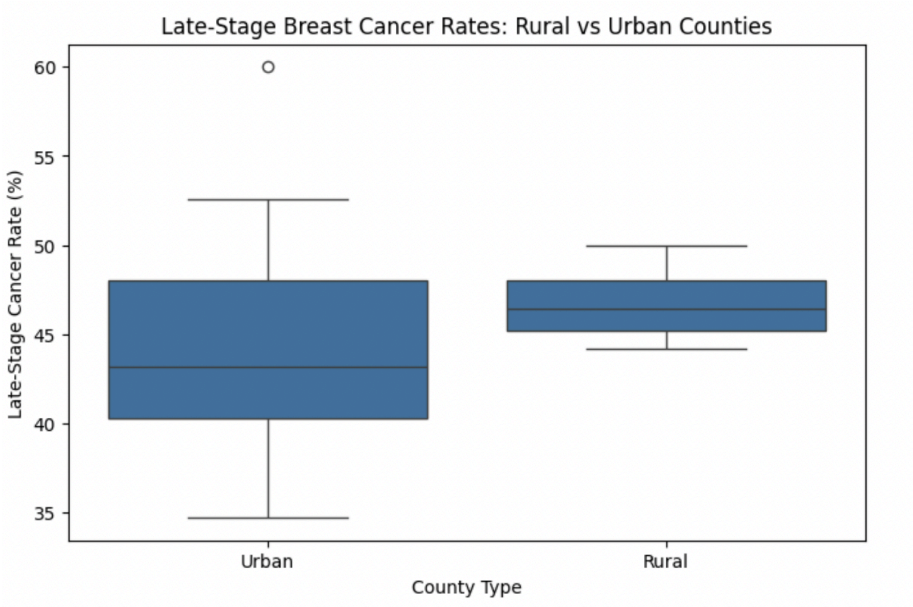
The relationship between uninsured rates and late-stage diagnoses appears weaker but still suggests that counties with more uninsured individuals may experience higher late-stage cancer rates.



The correlation heatmap supports these observations. Poverty rate shows a positive relationship with late-stage breast cancer rates, while median income shows a negative relationship. Poverty and uninsured rates are also strongly related to each other, reflecting broader socioeconomic patterns across counties.



Although the number of rural counties in the dataset was limited, the comparison suggests that rural counties may experience higher late-stage cancer rates. This pattern may reflect differences in healthcare access, highlighting the importance of improving screening availability and healthcare resources in rural areas.



Lastly, A linear regression model was also used to examine whether county-level socioeconomic factors could explain differences in late-stage breast cancer rates. The overall model was statistically significant (F-test $p = 0.0001$) and explained about 46% of the variation in late-stage cancer rates across counties ($R^2 = 0.464$).

Among the variables included in the model, poverty rate was the only statistically significant predictor ($p = 0.041$), indicating that counties with higher poverty levels tend to have higher percentages of late-stage breast cancer diagnoses. Median household income and percent uninsured were not statistically significant predictors in this model.

Conclusion

Overall, these findings suggest that county-level socioeconomic conditions, particularly poverty, may be associated with differences in breast cancer detection across California counties, which is expected since economic disadvantage can limit access to healthcare and early screening.

However, this analysis has several limitations. Some rural counties had suppressed cancer counts due to small populations, which resulted in missing values and an imbalance between rural and urban observations. In addition, limitations in the available datasets made it difficult to reliably analyze long-term trends in breast cancer rates among women under 50.

As a result, the analysis focused on cross-sectional relationships between socioeconomic factors and late-stage breast cancer rates. Future research could use more complete or longitudinal data to better understand geographic disparities. Expanding access to cancer screening and healthcare resources, particularly in economically disadvantaged communities, may help support earlier detection and improve health outcomes.